



Analyzing & Optimizing New York City Taxi Operations

VIVEK RAMDAS

Background - New York City Taxi Operations

New York, often called New York City (NYC) is the most populous city in the United States. New York is a global center of finance and commerce, culture, technology, entertainment and media, academics and scientific output, the arts and fashion, and, as home to the headquarters of the United Nations, international diplomacy.

Taxis play a crucial role in New York City's urban transport network. With the city's dynamic environment, taxi companies need to continuously adapt and optimize their operations to meet changing demand patterns, ensure profitability and enhance customer satisfaction.

Business Objective:

- To use the 2023 taxi trip data to uncover insights that could help optimize taxi operations.
- To analyze patterns in the data for taking informed strategic decisions to improve service efficiency, maximize revenue and enhance passenger experience.

Dataset Overview

Context:

The yellow taxi trip records include fields capturing details of the trips taken everyday by yellow taxis in New York City. The data is stored in Parquet format (.parquet). The data set is from 2009 to 2024. However, for this analysis, we will only be focusing on the data from 2023. The data for each month is present in a different Parquet file. The data was collected and provided to the NYC Taxi and Limousine Commission (TLC) by technology providers such as vendors and taxi-hailing apps and is publicly available.

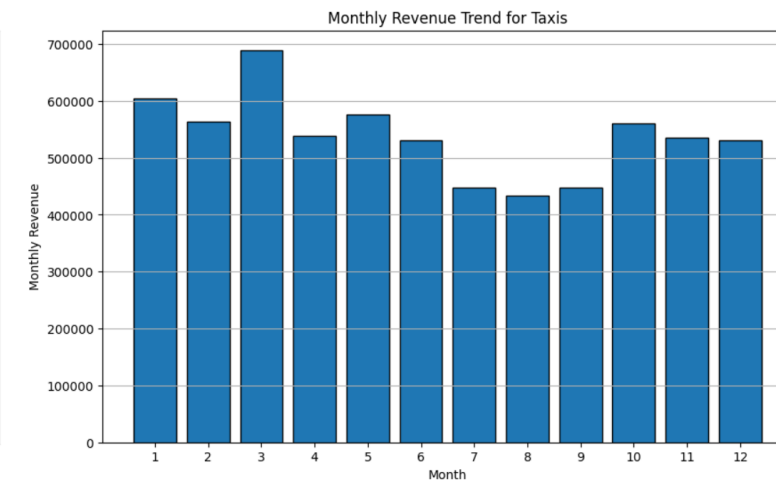
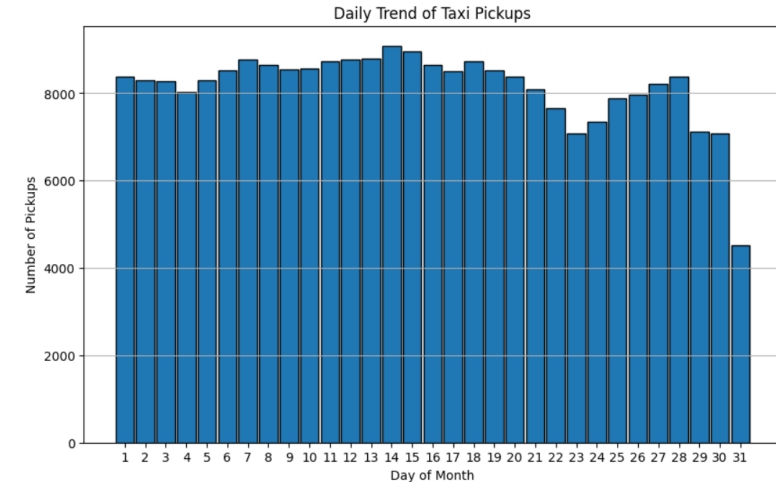
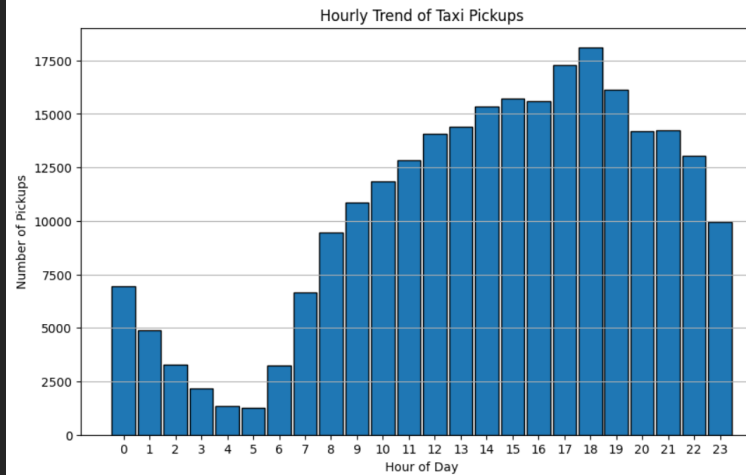
Content:

- The dataset contains a set of 12 Parquet files having records on attributes such as vendor ID, time and locations of pick-up and drop-off, payment type, fare amount and tips.
- Apart from the trip data, there is also access to a shape file containing geometric data about the taxi zones.

Preliminary Insights

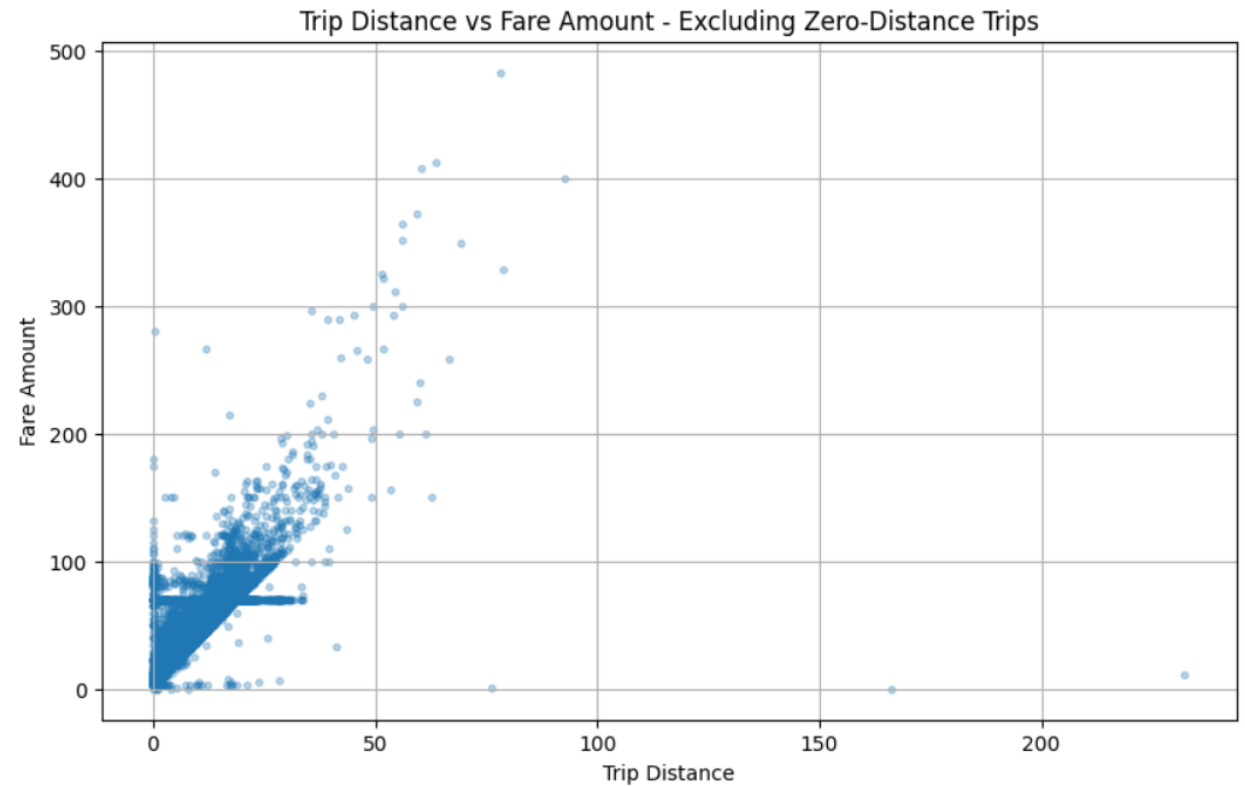
Datetime Analysis

- Busiest hours of the day: 17:00, 18:00, 19:00
- Sharp fall in hourly demand between 00:00 to 08:00
- Daily demand is evenly distributed (ignore dip on 31 as only some months have 31 days.)
- March has highest monthly demand.
- Drop in demand and revenue observed in Q3 (July, August, September)



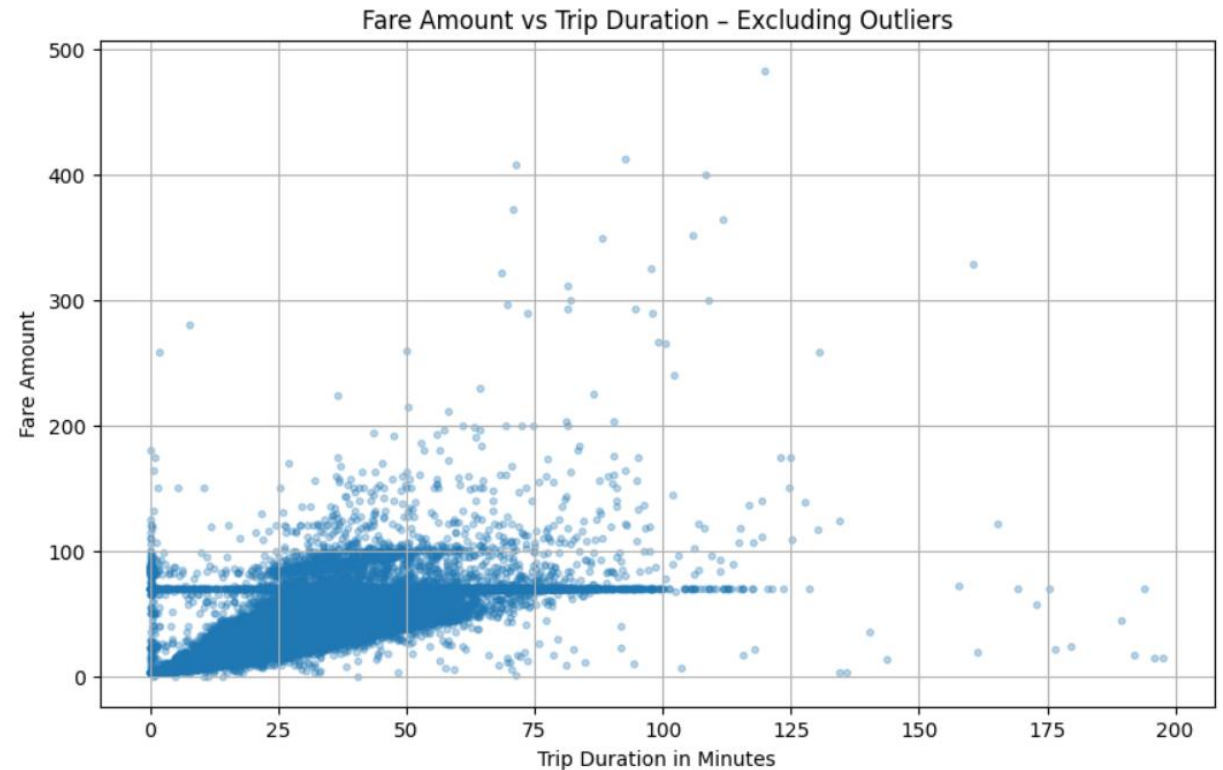
Correlation: Trip Distance & Fare Amount

- Linear Correlation.
- Fixed fare amount trips observed – charge of ~80 for a distance upto ~35 miles
- Limited spread of fare_amount across same distance values observed. This is likely depicting surge pricing during various time slots.



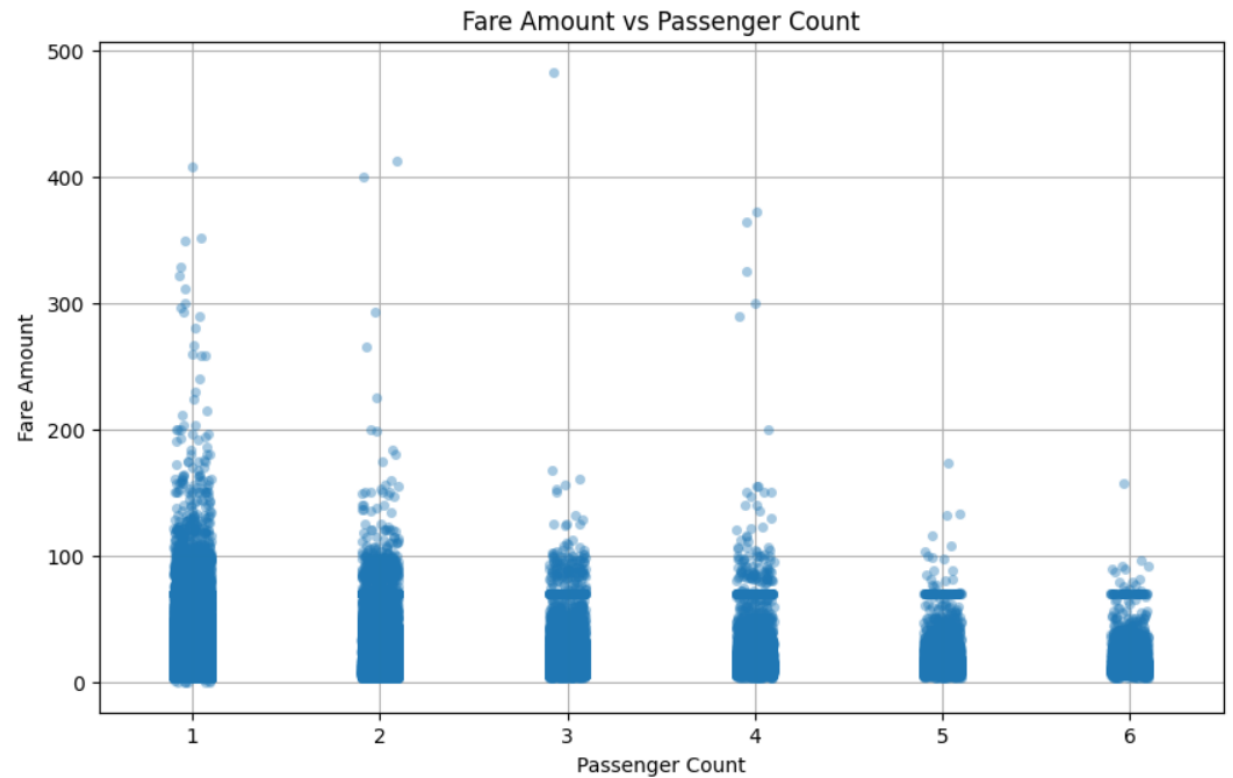
Correlation: Trip Duration & Fare Amount

- Linear Correlation.
- Fixed fare amount trips observed – charge of ~80 for a duration upto ~125 minutes
- Fair spread of fare_amount across same duration values observed. This is likely depicting surge pricing during various time slots.



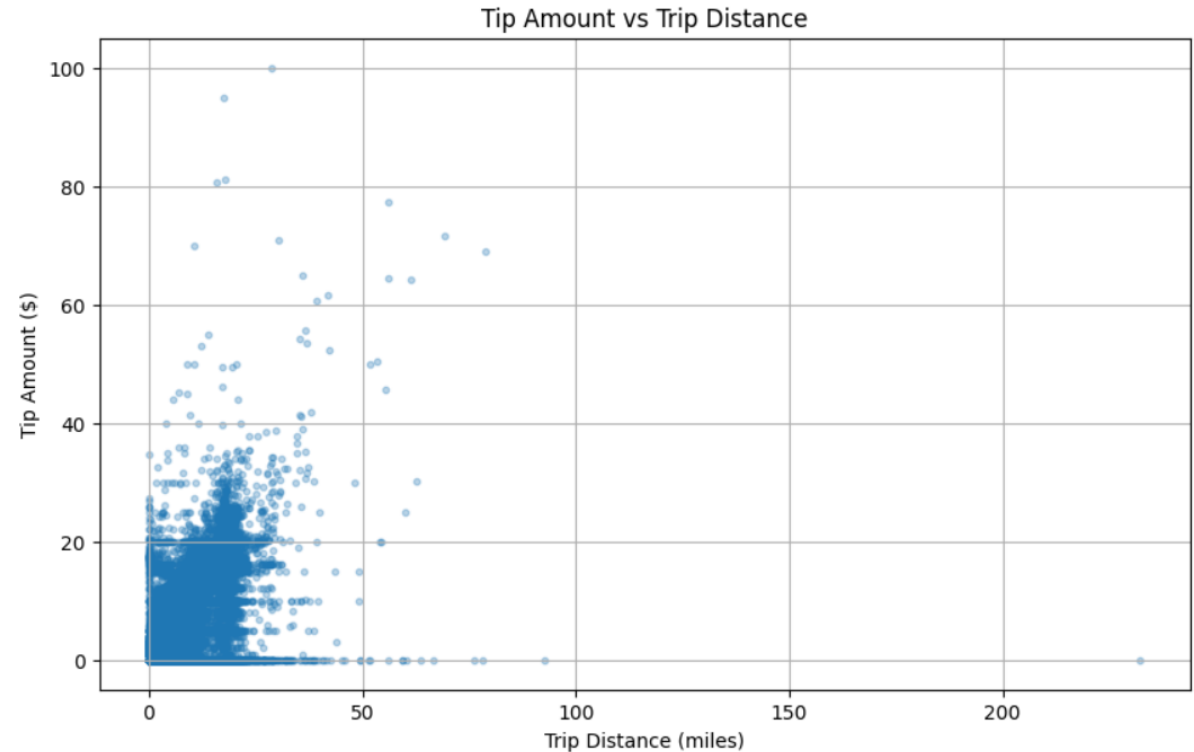
Correlation: Passenger Count & Fare Amount

- Minimal Correlation – almost independent.
- However, most datapoints are skewed towards 1 or 2 passenger trips.
- It is possible we have insufficient data points to arrive at conclusions.



Correlation: Trip Amount & Trip Distance

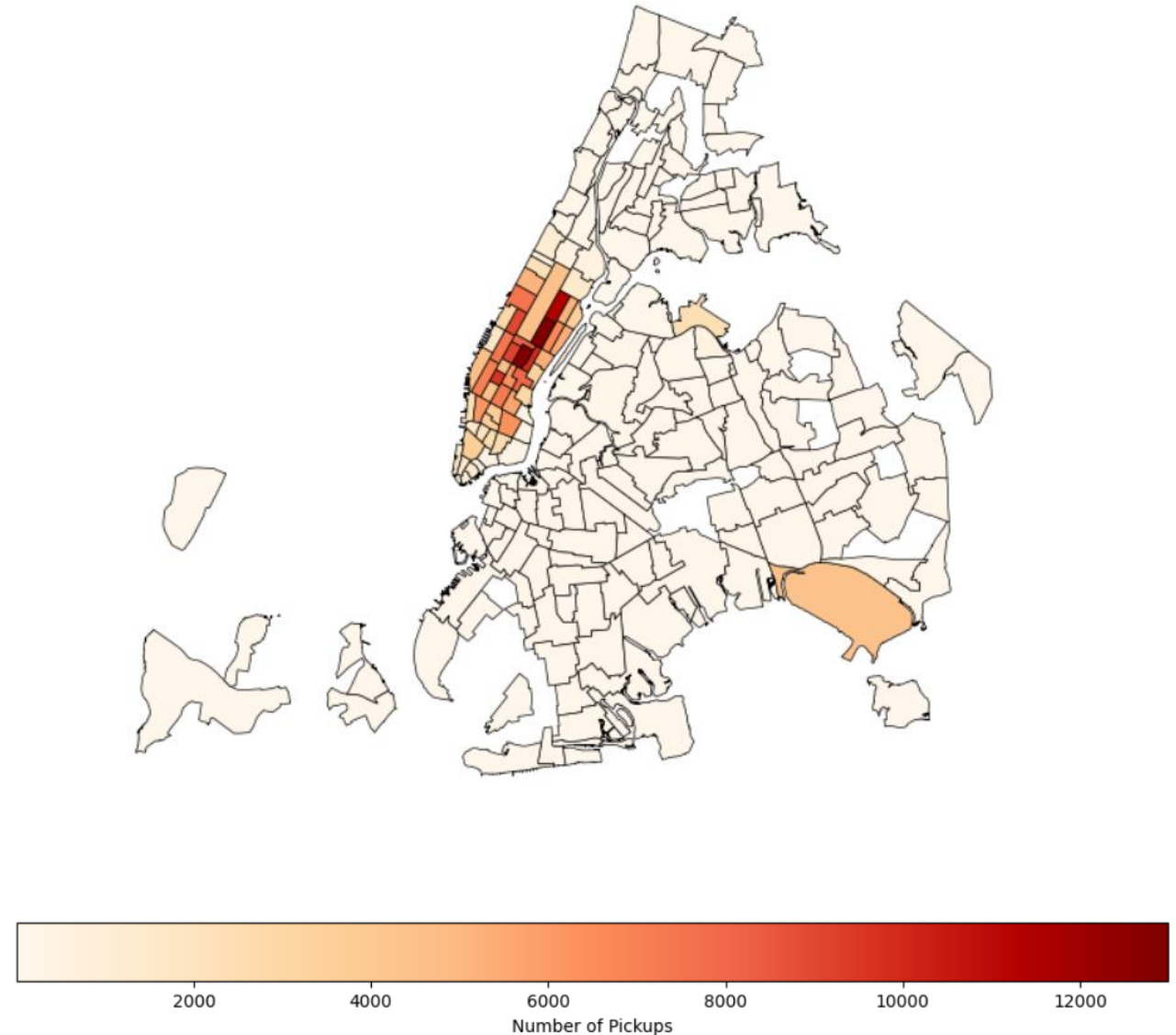
- Tip amounts clustered around short distance rides.
- It is possible that tipping declines when the fare_amount of the trip is high.



Correlation: Pickup Count and Zones

- Pickup counts were clustered around a small section of the city.
- Top 5 zones had PULocationID 237, 161, 236, 162, 186.
- Highest demand centered around a very small portion of the city - likely the business and/or commercial districts.

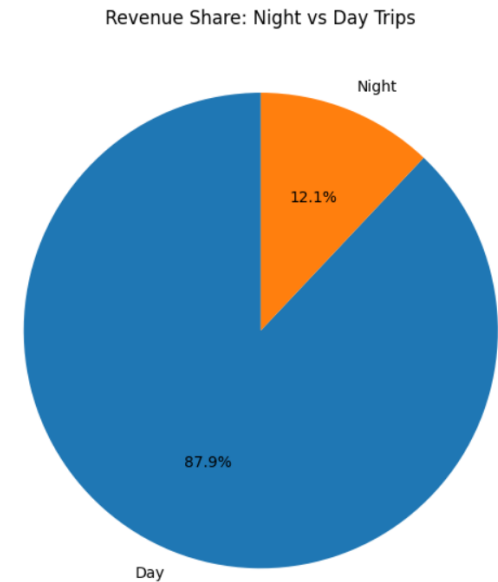
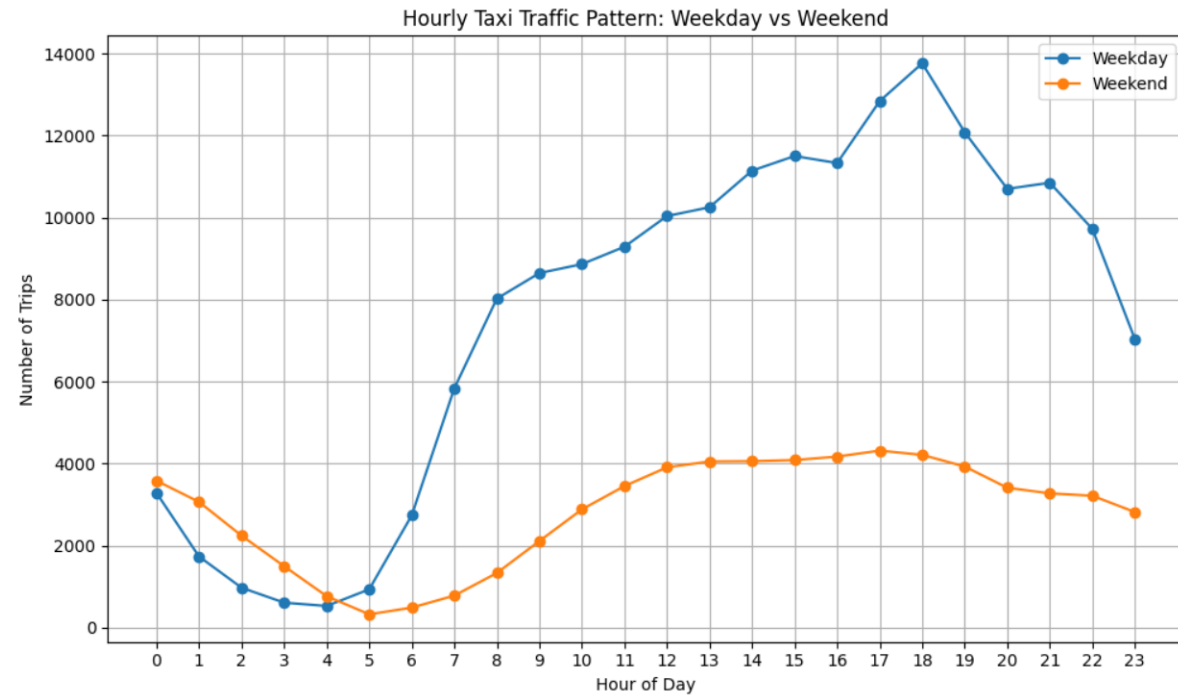
NYC Taxi Pickups by Zone



Detailed Insights

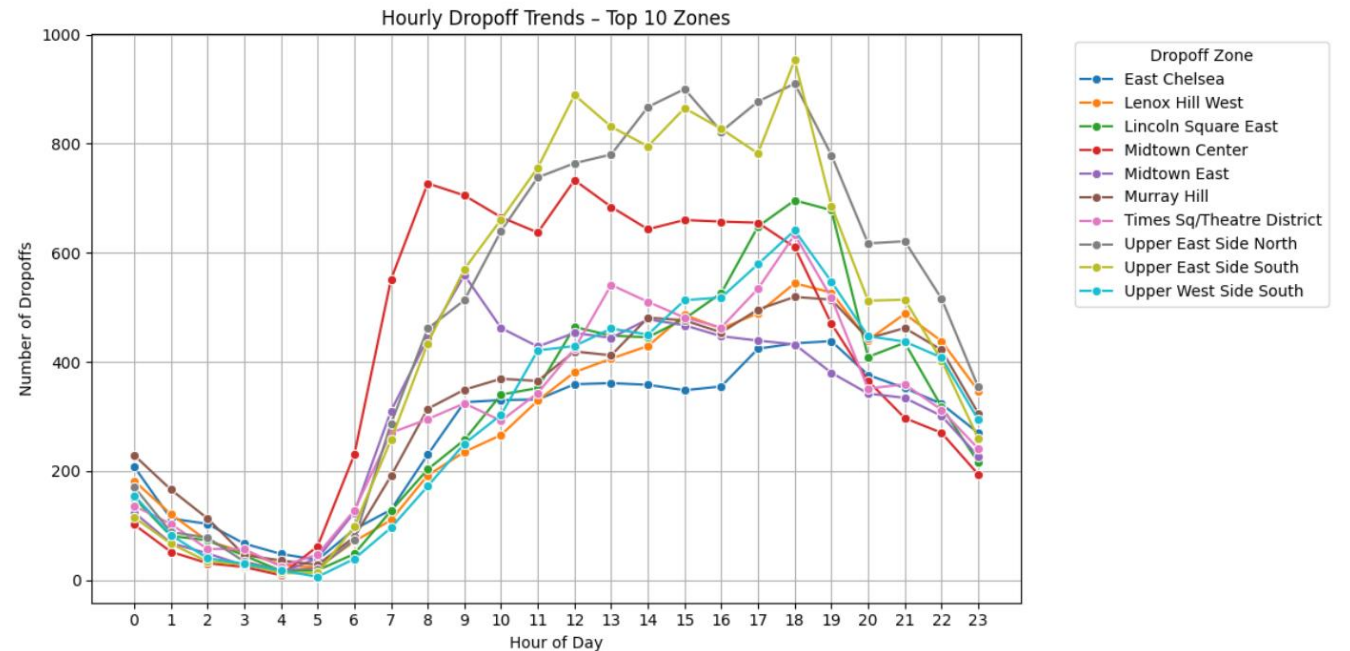
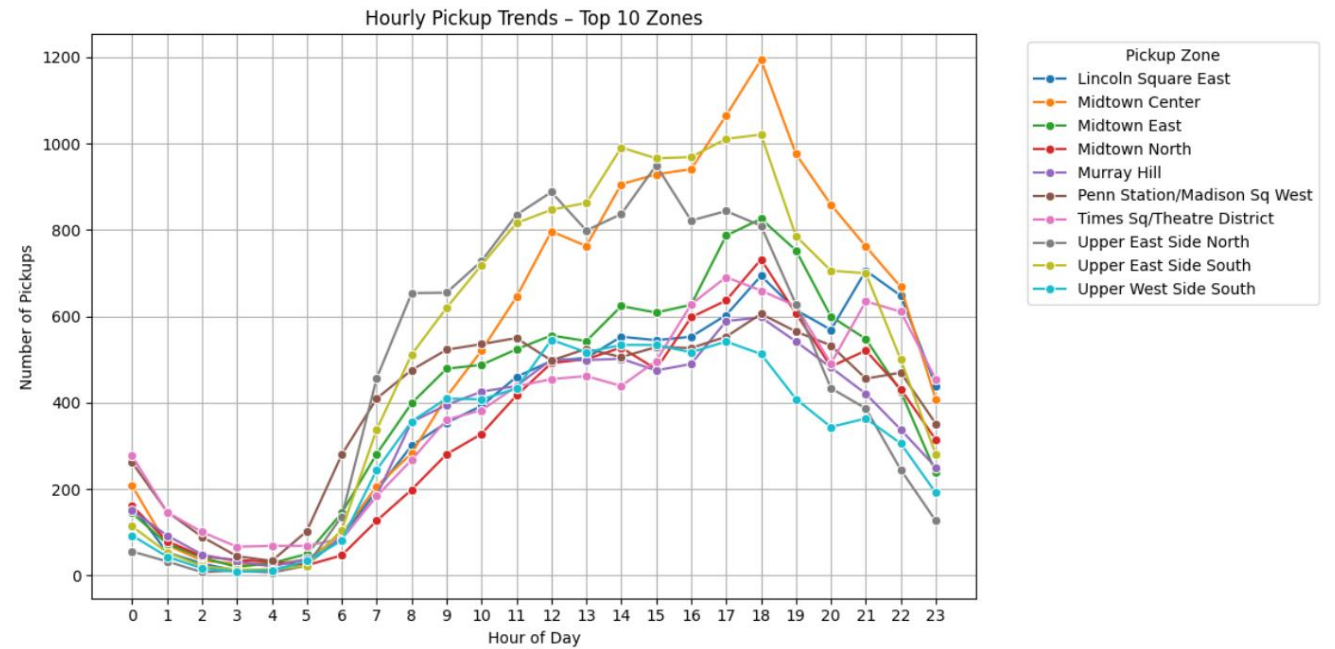
Datetime Analysis

- Busiest hours during weekdays were 17:00, 18:00, 19:00.
- Significant spike in demand starting from noon till late evening on weekdays.
- Demand on weekends remains almost equally spread across the greater part of the day. (Dip between 02:00 and 09:00)
- Slightly greater demand between 00:00 to 04:00 on weekends compared to weekdays.
- Day trips contributed to ~88% of the revenue.



Zonal Analysis

- Top 10 Zones as per hourly pickup trends are listed on the right.
- Top 10 Zones as per hourly dropoff trends are listed on the right.



Zonal Analysis

- Top 10 Zones as per pickup/dropoff ratio have been identified.
- Bottom 10 Zones as per pickup/dropoff ratio have been identified.
- Top 10 Zones as per night trip count have been identified.

Top 10 Zones as per ratio

	zone
65	East Elmhurst
118	JFK Airport
175	Penn Station/Madison Sq West
239	Willeys Point
103	Greenwich Village South
40	Central Park
236	West Village
151	Midtown East
150	Midtown Center
152	Midtown North

Bottom 10 Zones as per ratio

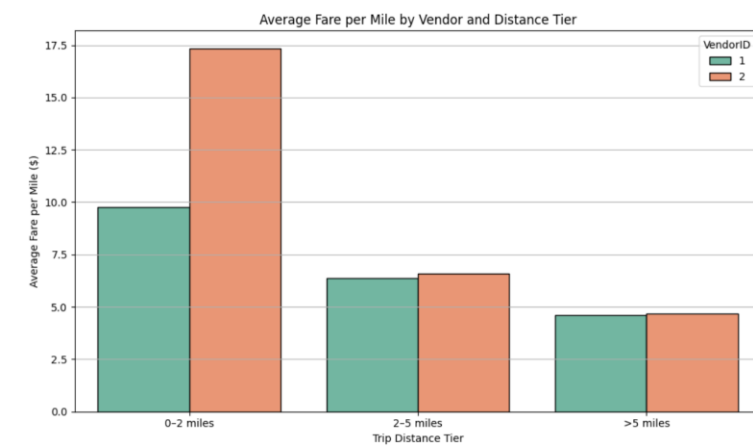
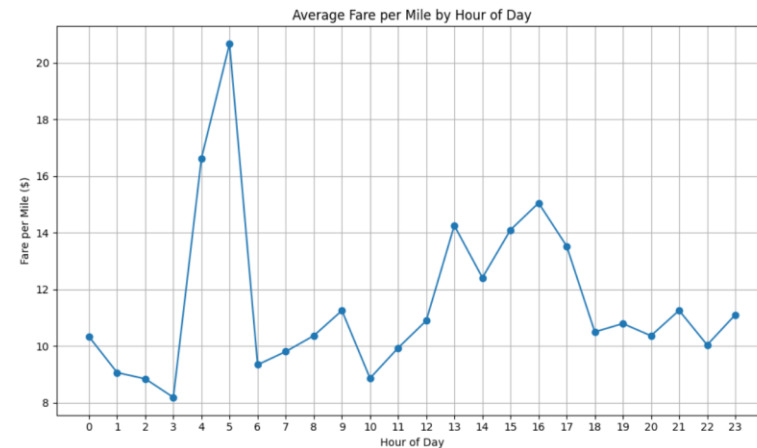
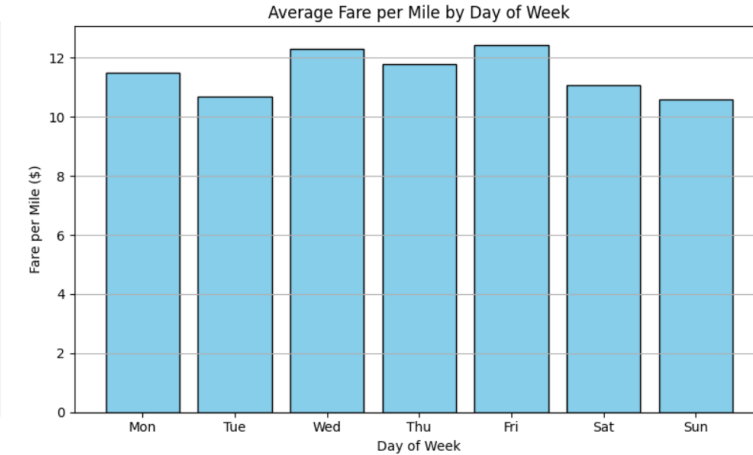
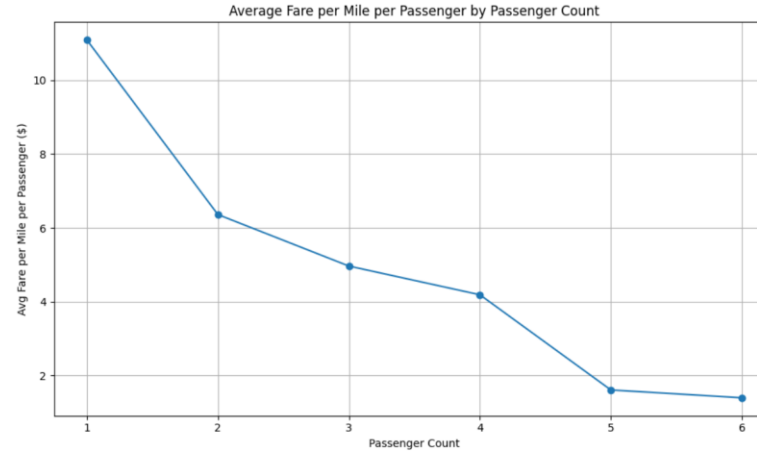
	zone
0	Allerton/Pelham Gardens
91	Forest Park/Highland Park
89	Fordham South
79	Eltingville/Annadale/Prince's Bay
76	Eastchester
68	East Flushing
222	University Heights/Morris Heights
62	Dyker Heights
228	Van Cortlandt Village
58	Cypress Hills

Top 10 Zones as per night trips

	zone
70	East Village
223	West Village
40	Clinton East
130	Lower East Side
205	Times Sq/Theatre District
59	East Chelsea
96	Greenwich Village South
92	Gramercy
166	Penn Station/Madison Sq West
145	Midtown South

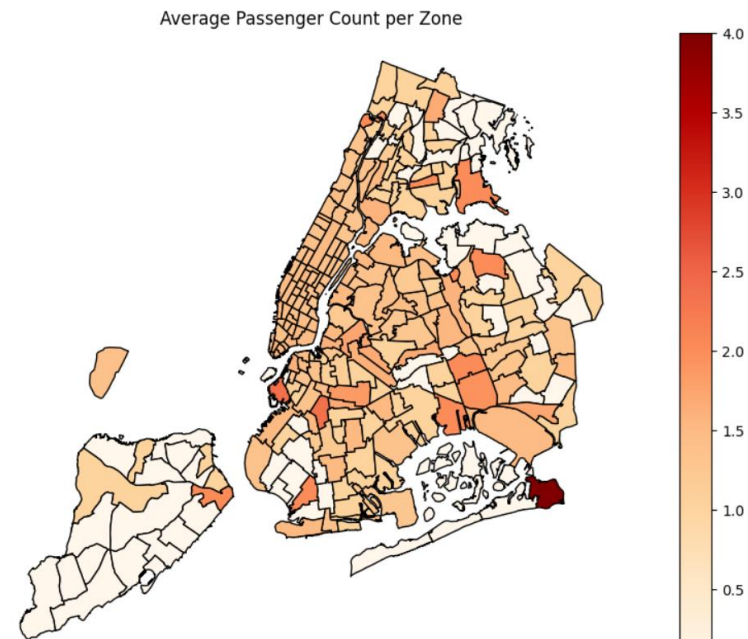
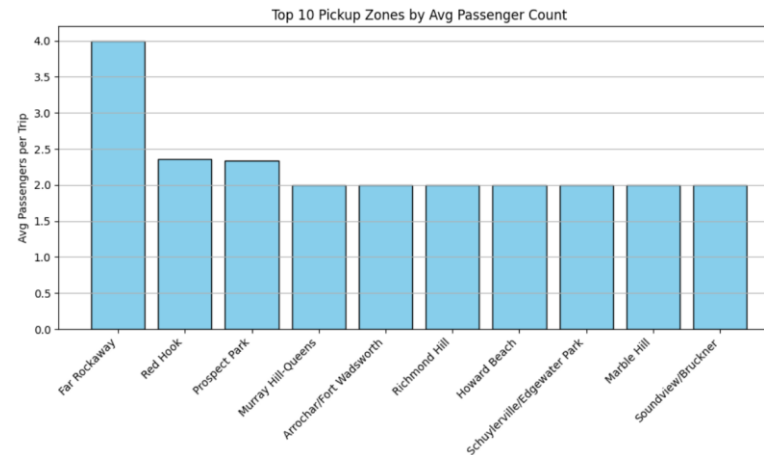
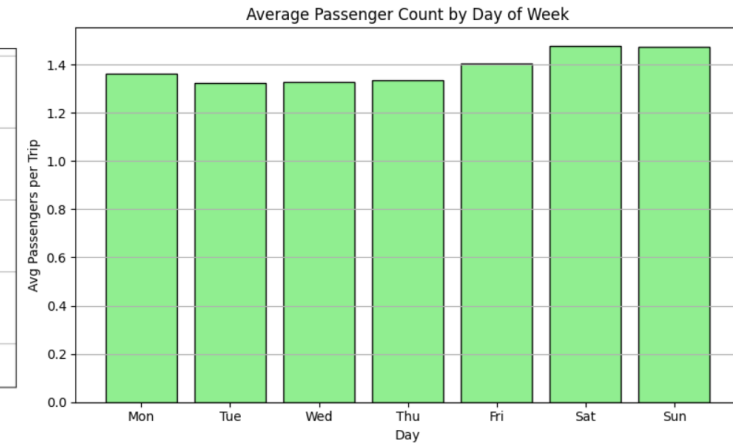
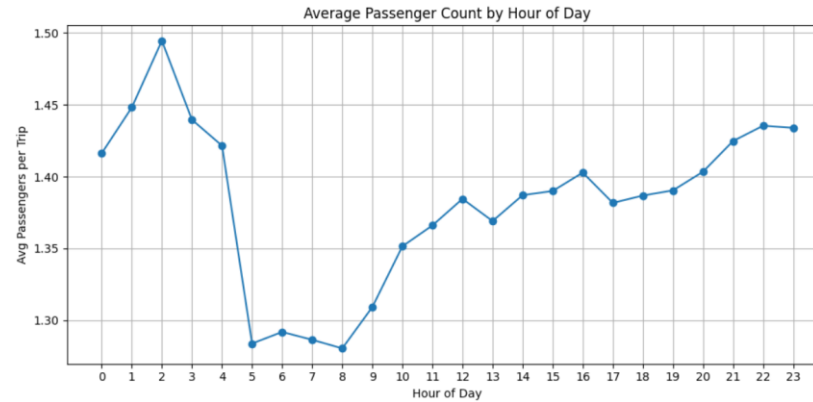
Average Fare

- Average Fare per mile has an inverse relationship with passenger count.
- Average Fare Per mile remains constant across weekdays and weekends.
- There is a significant spike in average fare during 04:00 and 05:00 hours.
- Vendor1 charges significantly lower Average fare per mile for shorter distances.



Passenger Count

- Passenger counts are higher during late night and early morning hours.
- Passenger counts remain same across all days.
- Trips with higher passenger counts appear to be centred around specific zones.
- The top 10 zones by average passenger count has been listed on the right.



Recommendations

Optimize Routing & Dispatching

1. Optimize night dispatch by increasing the deployments in regions with high night pickup count.
2. Concentrate deployments in zones where the pickup count is high. Top 5 zones are 237, 161, 236, 162, 186.
3. Optimize the deployments in alignment with the hourly pickup trends per zone. Follow the Top 10 zones graphs for guidance.
4. Increase the number of deployments to zones where pickup/drop_off ratio is high.
5. Optimize deployment of larger vehicles by restricting to zones where the average passenger count per zone is higher.

Optimize Deployments - Time, Day & Month

1. Reduce the number of overall deployments between 00:00 and 06:00. However, optimize by increasing the deployments in regions with high night pickup count.
2. Reduce the number of overall deployments on weekends.
3. Reduce overall cab deployments in Q3.
4. Reduce the number of overall deployments between 00:00 and 06:00.
5. Increase the number of overall deployments between 16:00 to 20:00 on weekdays.

Adjustments To Pricing Strategy

Following inclusions to be evaluated and introduced into the pricing strategy:

1. Weekend Surcharge per mile.
2. Extra Passenger Surcharge per additional passenger.
3. Late night surcharge between 00:00 to 06:00.
4. Zone based surcharges if feasible.
5. Demand Based Dynamic Pricing - increase earning potential of peak hours.
6. Standardization of price per mile across vendors particularly for short distances.

Thank You.